Telkomsel
by Telkom Indonesia

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Faculty of Economics and Business, Universitas Airlangga

ECO-CONNECT: REIMAGINING TELKOMSEL'S CONSUMER TOUCHPOINTS

Driving Digital Consumer Resilience, Green Monetization, and Next-Gen Experience Hubs

A Strategic Response to Telkomsel's "Green Self-Service: Digital Consumer Services Without Visiting GraPARI"

SUBMITTED BY: TIM DOA IBU

Team Leader: Lead Strategist

Team Member 1: Financial Modeler

Team Member 2: Technology Architect

Faculty / Major: Economics & Business / Management

Institution: Universitas Airlangga / Indonesia

Contact: contact.timdoaibu@gmail.com

Grand Theme: Future-Proofing The Economy: Digital Resilience and Inclusive Growth in Emerging Markets
Sub Theme: Leveraging Inclusive Technology to Drive Sustainable Economic Transformation
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EXECUTIVE SUMMARY: THE ECO-CONNECT STRATEGY

1. Strategic Context & Root Cause Diagnostic (30% Rubric)

The Squeeze: Telkomsel leads Indonesia (159M cellular & 8.5M broadband users) but faces a structural **Monetization Squeeze**: annual data traffic surges +25%, yet blended ARPU stagnates at Rp49,000 and market share slipped to 44%. Simultaneously, ±350 GraPARI branches log **15M walk-in visits/year**, generating massive OpEx (Rp375B/yr @Rp25k/serve) and **37,500 tons CO_2e** (2.5 kg CO_2e/visit, driven by customer travel).

Core Problem Metrics:
- **Traffic vs. ARPU:** Traffic +25% vs. ARPU +3%.
- **GraPARI Cost:** 15M visits × Rp25k = Rp375B/yr.
- **Carbon Impact:** 15M visits ≈ 37,500 tons CO_2e.
- **Adoption Friction:** 60% eligible for digital, but blocked by literacy & e-KYC.

Core Question: How can Telkomsel scale **MyTelkomsel** and Azure OpenAI-powered **Veronika** to migrate routine GraPARI visits into inclusive self-service, lifting ARPU while cutting emissions toward its 2030 target (-20% carbon intensity)?

2. The Solution Idea: Three Synergistic Pillars of ECO-CONNECT (45% Rubric)

Pillar	Strategic Initiatives & Operational Mechanisms	Impact & Value Moat
Pillar 1: Agentic Veronika & Inclusive e-KYC	• **Vernacular Voice AI:** Conversational agent in Bahasa and dialects (Javanese, Sundanese) for seniors/rural users. • **WhatsApp-Native Engine:** Full self-service without requiring app downloads, lowering entry friction. • **Biometric Remote SIM Swap:** Dukcapil-backed liveness e-KYC enabling instant eSIM delivery and lost-SIM courier dispatch.	• Migrates **7.5M visits** to digital by Year 3. • Resolves GraPARI's #1 walk-in driver. • Champions **SDGs 9 & 10**.
Pillar 2: Green Eco-Pass & AI VAS Engine	• **Telkomsel Eco-Pass:** In-app carbon counter tracking avoided travel emissions (2.47 kg CO_2e / visit), earning **Eco-Poin** for data vouchers & green merchants. • **Monetized AI VAS (Veronika CyberGuard):** B2C scam/phishing filter & family digital shield priced at Rp10,000/month. • **B2B ESG Syndication:** Anonymized mobility carbon savings syndicated into corporate client ESG reporting suites.	• Unlocks **Rp306B new revenue** by Y3. • Raises active ARPU by **+Rp3,000**. • Fosters brand advocacy (**SDGs 12 & 13**).
Pillar 3: GraPARI NextGen: Experience Hubs	• **24/7 Smart Kiosks:** Exterior automated biometric kiosks for physical SIM issuing, downsizing branch transactional space by 60%. • **MSME Digitalization Lounges:** Indoor space repurposed for 5G/IoT smart living demos and local enterprise digital training. • **High-Touch Assisted Care:** Staff re-skilled as digital ambassadors for complex disputes, fraud cases, and senior onboarding.	• Deflates GraPARI branch OpEx by **Rp172.5B/year**. • Reinvents GraPARI from cost drain into experiential retail hub.

3. Technical Architecture & Ecosystem Integration

ECO-CONNECT leverages existing Telkomsel investments: (1) **Azure OpenAI Service** orchestrates autonomous intent routing and tool execution via Core BSS/OSS APIs; (2) **Dukcapil e-KYC** enables real-time biometric liveness and NIK matching for bank-grade SIM security; (3) **FMC Unified CRM** pairs mobile and IndiHome subscriber data for personalized cross-selling.

4. Environmental Impact Methodology (GHG Protocol Scope 3)

Using the **GHG Protocol Corporate Value Chain (Scope 3) Standard**, avoided emissions are quantified by comparing physical customer journeys against digital interaction cloud footprints:

$$\mathrm{E}_{\text{avoided}} = \sum_{t=1}^{3} \text{Shifted Visits}_t \times \left(e_{\text{travel}} + e_{\text{facility}} + e_{\text{paper}} - e_{\text{digital}}\right), \quad \Delta e = 2.47 \ \text{kg} \ \mathrm{CO_2e} \ / \ \text{visit}$$

- **Avoided Travel (e_{travel}):** Avg. 12.5 km round-trip via urban motorcycle/car mix (0.18 kg $\mathrm{CO_2e}$ / pkm) = 2.25 kg $\mathrm{CO_2e}$.
- **Facility & Paper ($e_{\text{facility}} + e_{\text{paper}}$):** Grid electricity (0.78 kg $\mathrm{CO_2}$ / kWh) + paper printing = 0.25 kg $\mathrm{CO_2e}$.
- **Cloud Inference (e_{digital}):** Azure OpenAI inference and transmission = −0.03 kg $\mathrm{CO_2e}$.
- **Net Decarbonization:** Shifting 2.7M (Y1), 5.0M (Y2), and 7.5M (Y3) visits cumulatively eliminates **37,544 tons of $\mathrm{CO_2e}$**, directly supporting Telkomsel's 2030 −20% emission intensity target.

Decarbonization Dividends:
- **Total Avoided Emissions: 37,544 Tons $\mathrm{CO_2e}$** (equiv. to 620,000 trees).
- **Paperless Impact:** 15M paper receipts eliminated.
- **Societal Time Saved: 6.25 million hours** of travel and queuing eliminated (**SDG 8**).

5. Business & Financial Viability (3-Year Forecast)

ECO-CONNECT establishes a dual-engine model combining cost deflation with high-margin AI subscription revenue:

Financial Metric (IDR Billion)	Year 1 (2027)	Year 2 (2028)	Year 3 (2029)
Shifted Physical Visits (Millions)	2.7M	5.0M	7.5M
GraPARI Cost-to-Serve Savings (@Rp23k net/visit)	Rp 62.1	Rp 115.0	Rp 172.5
New Monetized AI VAS (Veronika CyberGuard @Rp10k/mo)	Rp 96.0	Rp 180.0	Rp 264.0
Eco-Pass Merchant Sponsorship & Brand Partnerships	Rp 15.0	Rp 28.0	Rp 42.0
Total Gross Economic Benefit	**Rp 173.1**	**Rp 323.0**	**Rp 478.5**
CapEx (AI voice fine-tuning, Dukcapil API, Kiosks)	(Rp 42.0)	(Rp 18.0)	(Rp 12.0)
OpEx (Cloud inference, marketing, staff training)	(Rp 25.0)	(Rp 22.0)	(Rp 20.0)
Net Cash Flow (NCF)	**Rp 106.1**	**Rp 283.0**	**Rp 446.5**
Cumulative Net Benefit	**Rp 106.1**	**Rp 389.1**	**Rp 835.6**

Investment Returns: NPV (10% WACC) = Rp 672.4 Billion | IRR = 142% | Payback = 7.8 Months. Cash flow positive in Year 1 while boosting active user ARPU by **+Rp3,000**, closing the monetization gap.

6. Execution Roadmap, Governance & Support Needed

Phased Rollout & Governance:
- **Horizon 1 (M1–M6):** Pilot WhatsApp SIM swap & 30 Jabodetabek kiosks; fine-tune vernacular voice AI with cross-functional agile squads.
- **Horizon 2 (M7–M18):** Deploy 350+ kiosks nationwide; launch Eco-Pass & Veronika CyberGuard; downsize branch manual desks.
- **Horizon 3 (M19–M36):** Complete MSME Lounge transition; syndicate B2B Scope 3 carbon credits to corporate clients.

Risk Mitigation Matrix:
- **Biometric Fraud:** Multi-factor liveness + telco SIM lock.
- **Elderly Resistance:** Staff re-skilled as Digital Concierges.
- **AI Hallucinations:** Deterministic guardrails + 30s human fallback.
- **Support Needed:** Dukcapil API sandbox & CMO sponsorship.

Conclusion: ECO-CONNECT transitions Telkomsel from an infrastructure-heavy telco into a high-margin, carbon-negative Digital Telco (DigiCo). By converting routine friction into decarbonization and generating **Rp835.6B in net cash flow**, Telkomsel future-proofs Indonesia's digital economy for inclusive and resilient growth.

APPENDIX & REFERENCES

Appendix A: GHG Protocol Scope 3 Emission Factor Assumptions

Parameter	Value	Source & Methodological Basis
Avg. Customer Roundtrip Travel	12.5 km	Based on urban GraPARI catchment area radius (BPS & Google Mobility index).
Urban Transport Emission Factor	0.18 kg CO_2e/pkm	Weighted average of 70% motorcycle (0.11 kg) and 30% private car/taxi (0.34 kg) (IPCC).
Electricity Grid Emission Factor	0.78 kg CO_2/kWh	Ministry of Energy and Mineral Resources (ESDM) Java-Madura-Bali grid emission factor 2024.
Outlet Energy per Interaction	0.25 kWh	Allocated branch HVAC, lighting, and computing power across average daily footfall.
Digital Interaction Cloud Carbon	0.03 kg CO_2e	Microsoft Azure Sustainability Calculator for OpenAI GPT-4 inference & edge network delivery.
Net Avoided Carbon per Shift	**2.47 kg CO_2e**	**Physical baseline (2.50 kg) minus Digital self-service footprint (0.03 kg).**

Appendix B: Scenario & Sensitivity Analysis (Year 3 Net Economic Benefit)

Scenario	Shifted Visits	Net Cash Flow (Y3)	3-Year NPV (@10% WACC)
Bear Case (-25% Adoption)	5.6 Million	Rp 321.4 Billion	Rp 468.2 Billion
Base Case (Expected)	**7.5 Million**	**Rp 446.5 Billion**	**Rp 672.4 Billion**
Bull Case (+20% Adoption)	9.0 Million	Rp 548.8 Billion	Rp 834.1 Billion

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